

Software engineering experience

- Describe a skill or knowledge you acquired recently that has been impactful for you. Why did you make this investment? What has the outcome been?

Over the past year, i had multiple instances where i got to know about something, or was given a task where a specific skill was required. So, i had to learn the skill. For example at observer technologies, i was introduced to multi tenancy database architecture, where you're always thinking about data isolation and making sure one customer's operations don't effect another's, i never heard of it because back then in course projects we always had one single database. But when i was given a task to implement an audit logging system, i learnt about it and how every API did go through a middleware for authentication. Talking about APIs, I got much more serious about API design, specifically with how to follow OpenAPI standards, versioning, and writing APIs so that every other developer can understand what is implemented. I learned about the different environments; dev, staging and prod. How we use Git and we push are changes or how we create and merge a PR. Back then i was very unorganized, and always pushed changes to master branch, which resulted in conflicts. It helped me now to ship APIs which are better documents, easier to test and understand, and require less back and forth which previously took a lot time in debugging.

- What new skill would you like to learn? Why do you think this is important or timely or interesting? Why do you think you will be good at it?

I want to learn a number of things. First, how to efficiently use AI tools. Because as fast as AI tools do get the things done, it also screws it. With that i can then dive into AI automation, and how i can automate my tasks through multiple agents. I do have a claude subscription, but i dont think i am getting most out of it except reaching the token limit.

I also want to learn low level systems programming. specifically Rust and Golang. Most of my background is high level. I did work on an on-prem multi orchestrator vulnerability scanner in Golang at my last internship, but it was all through AI with some basic understanding. I understand what's happening at the OS level in theory but I haven't had to own it in code. I like to work on Rust because i have taken courses on it to learn and i feel it forces you to think about memory explicitly without giving you the chaos of C. I spent time learning how PostgreSQL query planning works, how Docker networking actually functions, so i think that curiosity tends to transfer.

- What kinds of software projects have you worked on before? Which operating systems, development environments, languages, databases?

Most of my work sits at the backend and infrastructure layer. I prefer working on MAC, because its fast, secure, and provides a Unix-like environment that's very close to production Linux servers, which makes development and deployment smoother. The development environment i worked on vscode, cursor. Languages i worked on is python, nodejs, typescript, javascript, c++, django. Databases i worked on so far are mysql, postgresql and mongoDB.

Beside my course projects softwares, especially at internship, at Observeri Technologies, i worked on a cybersecurity GRC tool for enterprises. I worked on maintaining backend services in node.js / express.js,

Currently at Data Science Dojo, working on an enterprise AI agent platform. I work in the platform team, building and maintaining APIs, deploying models on Azure, managing the API Management Gateway (APIM) that sits in front of every customer's environment, and backend features.

On the academic side, in my final year proj, i worked on a a social impact mobile platform built with Node.js / Express (hosted on render) and PostgreSQL (hosted on neon db) , 40+ tables, messaging via Socket.io, google OAuth, otp through AWS SES, push notifications via Firebase FCM, file uploads using uploadthing, and AI event recommendations using Recombee.

- Would you describe yourself as a high quality coder? Why? When you are looking at code, how do you decide if it is high quality?

Honestly, at this stage, i wont say i am a PRO in writing code. Infact i dont even write code by hand on a vscode editor, specially after AI. so, i'd say i write decent code but more than that i know enough to recognize whats wrong when things fail. For me, good code is not if its working or if its efficient (AI takes care of it already very well) but when my teammate can pick it up without me explaining it to them. When I look at someone else's code, the first thing I ask is whether it's readable, are the standards defined followed, or its all unstructured. Then I look at error handling. Most of the time, we only think about the success scenarios, and test them well, but what about when the code fails, code looks fine until something unexpected happens and then we are clueless, because we dont know the error, and its now impossible to debug.

- Outline your thoughts on open source software development. What is important to get right when you are working on open source projects? What open source projects have you worked on? Have you been an open source maintainer, and can you point to those projects?

For me a good open source project is its documentation and contributor experience. writing tons of code is very easy these days. If you cant run it locally in under 10 minutes, most people will just leave. (Speaking from personal experience, this is where i suffer always when downloading a open source proj)

I haven't been a maintainer of any open source project yet. My github has academic and project work. I tried GSOC, spent two weeks or so exploring the repositories but due to my university work load, I couldn't continue and contribute. That's something I want to change, and this position is honestly the best chance for me to do it and a big reason why this role interests me.

- Our primary languages are Python, Golang, C/C++, Rust, JavaScript / TypeScript, and Bash. Please write about your familiarity with any of these, prioritizing the ones you know best. How extensive is your experience of them? How do you test applications in these languages?

I love working with python, and JavaScript / TypeScript. Mainly because i am working in these languages right now and i have my most experience in it. I cant pick one between python, and JavaScript. Python has much more support, at my current position i build apis in python with django as the framework, but for me javascript is easier because i started with it a c++/c i used in my course of OOP extensively when learning about the concept of classes, so yeah i have a good understanding about it as well.

Bash I use regularly for scripting, automating build steps, writing GitHub actions workflows, and general navigation through the terminal on my MAC. I'm comfortable at the command line but I wouldn't call myself a Bash expert. I know enough to get things done.

Golang, Rust are languages I haven't worked with in any real project yet. I understand the concepts and I've read enough to know what they're suited for.

- How comprehensive would you say your knowledge of networking is? How have you gained this knowledge?

I genuinely enjoy, learning and implementing Networking related concepts. Been using cisco packet tracers since ages. I took a dedicated course of Data Communications and Networking at Habib University which gave me a solid foundation and really piqued my interest in this field, we covered the full layered TCP/IP architecture from physical and data link layer protocols all the way up through network, transport, and application layers. We also studied the routing algorithms, packet switching, error and flow control. I also wrote a paper as a project for this course; Security at [Network Layer - IPSec and VPNs](#)

- How have you used relational data, such as SQL databases, in your work? How much data model design work have you done? What has been unexpected for you in working with structured data models and design?

Yes, i have been using SQL extensively in almost all my course projects and internships. I designed a schema with 30+ tables that involved understanding the database concepts, and a lot of decisions about normalization, foreign key relationships, and what to index.

One thing which is really unexpected sometimes is the foreign key constraint queries, that occur when someone else changes the schema, but does make sure to correct it everywhere it is being passed. For me, understanding the problem domain you're modelling, that is the hardest task. With that in hand you layout the main entities/process and from that create a basic model. Then keep reiterating/normalizing this model into more fine tuned versions until you have a well defined schema.

- Which NoSQL or unstructured data stores have you used? Are there any you would not use again, and why? How do you reason about choosing the right data store for a project?

i have used mongoDB. I used it on the AI Study Buddy project. I wouldn't use nosql until the data is explicitly and entirely not relational OR i get the advantage of the performance benefits of nosql AND it's worth the cost of losing ACID transactions, if your use case has data structures

that fit the retrieval model of a particular nosql database. One example; they are bad at aggregations because of the high cost to retrieve the document just to aggregate on one field.

- How comprehensive would you say your knowledge of a Linux distribution is, from the kernel up? How familiar are you with low-level system architecture, runtimes and Linux distro packaging? How have you gained this knowledge?

I would say my Linux knowledge is intermediate as i work every day in the linux environment. I fairly understand the hierarchy, and I can debug most things that go wrong. This knowledge mostly comes from my course Operating system which i took in my undergrad. I can write bash, troubleshoot and deploy Linux containers. Very good theoretical knowledge of Linux components and commands but my production experience with core Linux is limited.

- Describe any experience you have with low-level embedded systems engineering, on Linux or other embedded operating systems and code bases.

None professionally but I have a good knowledge of embedded development conceptually and I've worked with microcontrollers in my digital logic and design course where we made a foosball-like game on BASYS-3 FPGA board using Vivado. In another course of IoT and cloud computing, I worked on an ESP32 microcontroller to build a smart home system. The ESP32 read from temperature, humidity, light, and RFID sensors and communicated with AWS IoT Core over MQTT with TLS, so I have experience in these projects but I haven't done serious embedded work.

Your favorite or best personal project

Many candidates have tackled some personal project, not because it was coursework or curriculum, but because it was very interesting to them. This might be something from scratch, or it might be a fork or contribution to a piece of open source. Don't worry if you don't have one, you can just skip this section. But if you do, please answer these questions, and we might ask you to walk us through it later.

- Briefly describe this personal project. What were you hoping to achieve?

So it would be unjust if i just named one in my whole 4 year undergrad journey. So, ill pick the best 3 i worked on as they are very close to my heart and i learnt a lot.

The first is Kaarvan, my final year project. It's a mobile application that connects volunteers, donors, and NGOs in Pakistan. Our main goal is community building and connect people for a cause. The problem it solves is real: people in Karachi who want to attend events, volunteer or donate have no central place to find opportunities, and NGOs have no good way to reach them. We built 1-1/group messaging, event and communities discovery, fundraising.

The second is a legal judgment prediction we built and published as a research paper. The idea was to help a layman to understand the legal system and what charges he may face if he commits a crime. We trained transformer models on actual Pakistani criminal court judgments and see if the model could predict verdicts, with correct fines, charges, legal sections, when given a crime scenario and witness statements. No one had done this for Pakistani law before.

The third is IoT smart home. We built a full pipeline from an ESP32, which read temperature, humidity, light, and RFID, all the way through to a Power BI dashboard, using AWS IoT Core, MQTT over TLS, Lambda, S3, and Athena.

- What inspired you to write this piece of software?

Kaarvan came from living in Karachi. The gap between people who want to help and organizations who need donations and visibility. Also when i donated, i never knew where my donations go, I needed a platform which has transparency and track record.

The legal judgment project came was also on a similar idea because in Pakistan, the judicial system is totally on paper, difficult to understand, slow and inaccessible to most people. If a model could give someone a rough sense of what a court might decide before they even hire a lawyer, that would be very useful.

The IoT project started as coursework but became something more as i learnt a lot beside my software field, mainly how to interact with hardware.

- What language did you choose to write it in, and why?

Kaarvan: Node.js/Express for the backend and React Native for frontend to target both platforms with same syntax. PostgreSQL because the data was genuinely relational.

Legal judgment project: Python entirely because it is just the right tool for ML work. Most libraries are available in python.

IoT project: C++ for the ESP32 because that's what the Arduino ecosystem runs on. Python and SQL on the cloud side for data processing and Athena queries.

- Which part of the exercise did you find most difficult? What did you learn?

On Kaarvan, implementing messaging was harder than I expected. The message were not receiving real time, but only when you refresh the message page. Also, once this was solved, The socket id, which is probably what I would use to target a recipient, keeps changing (such as on refresh). I required constant debugging, learning in new ways and careful thinking about state that I hadn't dealt with in web work before.

On the legal project, the hardest part was building the dataset. Pakistan has no standardized legal database. We manually scraped pdfs from court websites, ran OCR to extract text, used few-shot prompting with Gemini 2.0 to extract five structured fields from each judgment.

- How did your code perform?

All of these projects, karavaan was the one which needs to be test under load. As it was supposed to be used by 1000-5000 concurrent users in the first phase. We did it with JMeter.

- Was there any particular part of this system that you felt a need to get absolutely, completely perfect? Why? How did you go about that?

I guess the perfection was needed in the legal system, our professor also emphasized a lot on the data quality. We were making claims about Pakistani law. Hallucination or wrongly extracted training data would invalidate everything. That's why we manually verified the dataset before touching any model.

- Is there anything you would do very differently if you were tackling this problem again now?

What we could do i (if we did not have any compute constraint), is that from the very start we built a larger and more diverse dataset, and experimented with bigger models. We were constrained by compute on the free version of google colab and couldn't fine-tune anything like llama-8B, which would likely have performed significantly better.

Education

We consider academic results in high school and university for all roles, regardless of seniority. In every discipline, from engineering to marketing to operations and sales, we intensely value colleagues who are able to puzzle through difficult problems and find the optimal path forward.

- How did you rank in your final year of high school in mathematics? Were you a top student? On what basis would you say that?

Out of the class of 15, I was in the top 3. I would say that on the basis of my maths course high school second year (grade 12) percentage which was 88%.

- How did you rank in your final year of high school, in your home language? Were you a top student? On what basis would you say that?

Due to covid, the exam of urdu language was not conducted. But just maths, physics, chemistry. But in my grade 10th, i secured 74%.

- Please state your high school graduation results or university entrance results, and explain the grading system used. For example, in the US, you might give your SAT or ACT scores. In Germany, you might give your scores out of a grading system of 1-5, with 1 being the best.

I completed my intermediate (high school) in Pakistan under the Pakistani board system, which grades on a percentage scale. I got 86%

- Can you make a case that you are in the top 5% in your academic year, or top 1%, or even higher? If so please outline that case. Make reference where possible to standardized testing results at regional or national level, or university entrance results. Please explain any specific grading system used.

Yes so through my school (grade 9th and 10th) i was in the top 5 of my class, and high school i was in my top 3 of my engineering cohort. Grading system used is the typical 100-point system. Also, to be selected at Habib University, I was in the top 300 out of 4000 students, And I also received a 100% merit scholarship via [HUTOPS](#) program at Habib University (only 5% students receive it per batch).

Pakistan's Secondary Grading Scale			
LETTER GRADE	NUMERICAL RANGE	DESCRIPTOR	WES EQUIVALENCY
A+	80–100	Exceptional	A
A	70–79	Excellent	A
B	60–69	Very Good	B+
C	50–59	Good	B
D	40–49	Fair	C
E	33–39	Satisfactory	C-
F	0–32	Fail	F

- What sort of high school student were you? Outside of class, what were your interests and hobbies? What would your high school peers remember you for?

Outside of class, I always tried various extracurricular activities happening at my school. I participated in robotics, swimming classes and horticulture for the 2 years of my highschool. I also was a part of my school band, where i was the chief flutist and led my flute group. Along with that, I was an active member of the school's disciplinary society as a prefect and in my final year of highschool, i was nominated as a house captain.

Outside my school, I was also in Scouts, where i participated in various voluntary services within my community and scouting camps.

But something that my peers would know me for was definitely volleyball and cricket. I was

always a sports freak and apart from athletics, I would often get selected to represent my school in interschool competitions for volleyball and cricket.

- Which university and degree did you choose? What other universities did you consider, and why did you select that one?

I was always sure about going into tech field, doing my bachelor's in computer science. I was a computer freak all my life, and getting to do that as a career was the only option for me. I was fortunate enough to be accepted in my dream school [Habib University](#), for my desired field. To cherry the cake, I was granted a 100% merit scholarship upon my entrance, for which I had to maintain a >3 CGPA throughout my degree.

- Overall, what was your degree result and how did that reflect on your ability? Please help us understand the grading system for your results.

At Habib University, for every course you'd be graded based on mid term and final exam, along with class participation, quizzes, lab work, presentation, and technical project submission. I made sure I closely meet all my submissions, appeared for quizzes, and studied hard for exams. I graduated my bachelor's with a CGPA of 3.33.

- During all of your education years, from high school to university, can you describe any achievements that were truly exceptional?

During my [highschool](#), I was selected to lead a student body house (a big achievement in my school). As a house captain, I instilled in myself leadership and communication skills. To be selected at Habib University, was itself a achievement because out of 3000 students, only 350 are selected per year. And I also recieved a 100% merit scholarship via [HUTOPS](#) program at Habib University (only 5% students recieve it per batch).

- What leadership roles did you take on during your education? Did you conceive of, and drive to completion, any initiatives outside of your required classwork?

As mentioned earlier, i served as a house captain in my grade 12th. Where I managed multiple schooling events, fests, sport days, science fairs along with the student body, teachers and management.

Apart from this, during my university, i also served as the treasurer of the computer science and engineering club, where we organized technical talks, gaming events, competitions. It was a very learning experience working with different stake holders.

Outside my education, in my community, i am also part of a students orgnization ([Toloba ul Kulliyaat ul Mumenoona](#)), where i am serving as a joint secretary in my neighboourhood.

- Did you participate in any social or community initiatives during your education? Were you recognized in any way for this?

As mentioned earlier, i also served in scouts, participating in multiple voluntary events (food distribution, plantation drives, beach drives, scouting camps) within and outside my community.

Since now i am in [Toloba ul Kulliyat ul Mumenoona](#), we mostly work in organizing events for students, like book bank (providing books at a lower cost to school students), short courses sessions, sporting events, short quizzes.

Context

- Outline your thoughts on the mission of Canonical. What is it about the company's purpose and goals which is most appealing to you? What do you see as risky or unappealing?

I couldn't find a formal mission statement for Canonical online, but it's clear from what they do, i.e. open source and free software. Ubuntu was the first time Linux on the desktop, and it was close to a real commercial experience. People started to talk about open source, which most people weren't having before. I want to work on a software that's accessible and free, Canonical has made it possible via open source without money. It really allows individuals to learn, develop, contribute, and work with these projects/softwares which is usually done in big tech companies. You see, Ubuntu is everywhere, anyone can access its source code, experiment and make a new operating system. The risk attached to it is that when your main product is free, and you are not able to monetize through it, it really is hard to balance the company without compromising the open source spirit.

- How should Canonical set about winning, commercially?

Start offering support and managing services. Most companies using ubuntu want to scale but don't want to manage it themselves, Canonical can start giving that support since we have specialised people for it. Canonical has Ubuntu Pro and Landscape, but I think there's more room to grow in the managed Kubernetes and AI infrastructure space.

- How should Canonical amplify its impact in open source?

Honestly, by making better the contributors experience, most of my friends in the university fears that the open source is difficult to understand, specially from the start of setting up. I feel it strongly. If someone wants to contribute and they can't run the project locally in under 10 minutes, they're gone. Canonical has the resources to make that seamless. Start from targeting students of early undergrad, Less than 10% of my university has ever worked on opensource, we need to take it as any other field of computer science. Sponsoring bootcamps, meetups is best to engage contributors, so that in future, these students can amplify the open source field.

- Why do you most want to work for Canonical?

Three reasons. First, I want to work on something that actually runs in production at scale, is accessible to anyone without any payment restriction, or unlike other software who just runs and used internally at one company. Second, I mentioned in my earlier answers that I never got to contribute to open source properly, but i always wanted to. This role will give me this chance practise this. I don't want to miss it again. Third, remote work. I started freelancing remotely back in 2019, so working independently and managing my own time is not something I'm trying

out, remote work only goes well when everyone actually cares about the output, not just the hours. That's the kind of environment I want to be in.

- Which Canonical products would you most like to work on?

Ubuntu itself and anything around the cloud and infrastructure layer. Given my current work with Azure, APIM, and building platform APIs, that's where I feel most useful right now.

- What gives you the most satisfaction at work?

The best satisfaction i get is when something i developed is used by others without me telling how to use it. To be able to write a self explanatory code, with a simple process to run and test it is a win win for me

- What would you most want to change about Canonical?

Honestly, hard to say when I haven't worked there yet. But, i liked the way the hiring process is structured for only the ones serious about this role will ultimately put an effort.

- What gets you most excited about this role?

The chance to work on low-level systems as most of my work has been at the application layer. Second, This would be my first role at a company fully dedicated to software, and honestly that's both exciting and a bit intimidating. I'm looking forward to learning from people who have spent their careers doing this. At the same time, I've picked up a pretty varied set of experiences across backend, infrastructure, and platform work, and I'd hope some of that is useful to the team too. It goes both ways.